TT-C-645b
November 14, 1967
SUPERSEDING
Fed. Spec. TT-C-645a
August 3, 1961

FEDERAL SPECIFICATION

CREOSOTE, COAL TAR,

TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type and three classes of creosote obtained from coal tar and used for the preservative treatment of timbers of different forms and species (see 6.1).

1.2 Classification.

1.2.1 Classes. Coal-tar creosote shall be of the following classes as specified (see 6.2).

1 - Min. sp. gr. at 38° C. 1.050;
min. distillate to 355° C. - 75 percent.

2 - Min. sp. gr. at 38° C. 1.050;
min. distillate to 355° C. - 65 percent.

3 - Min. sp. gr. at 38° C. 1.080.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-D-729 - Drums: Metal, 55-Gallon (for Shipment of Noncorrosive Material).
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" X 48" 4-Way (Partial) Pallet Skids, Runners or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations:

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are a part of the Code of Federal Regulations (Revised 1956) available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402. Orders should cite "49 CFR 71-90 (Rev. 1956).").

Federal Insecticide, Fungicide, and Rodenticide Act:

(Application for copies should be addressed to the U. S. Department of Agriculture, Agricultural Research Service, Pesticides Regulation Division, Washington, D. C. 20250.)

American Wood Preservers' Association (AWPA) Standards:

A1 - Standard Methods for Analysis of Creosote.
A4 - Standard Methods of Sampling Creosote Oil in Tank Cars.

(Application for copies should be addressed to the American Wood Preservers' Association, 839 Seventeenth Street, N. W., Washington, D. C. 20006.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

American Trucking Associations, Inc:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Material. Coal-tar creosote shall be a distillate derived entirely from tar produced by the carbonization of bituminous coal.

3.2 Properties. The properties of the coal-tar creosote shall comply with the requirements in table 1.

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Table I. Requirements of coal-tar creosote 1/

	Class 1		Class 2		Class 3	
	For pressure treatment of poles		For ties, lumber, structural timbers, land piles and posts, or for butt-treating poles		For piles, lumber, and structural timbers all used in coastal waters	
	Not less than	Not more than	Not less than	Not more than	Not less than	Not more than
Water, percent by volume	--	3.0	--	3.0	--	3.0
Specific gravity at 38° C. compared with water at 15.5° C.	1.050	--	1.050	--	1.080	--
Material insoluble in benzol, percent by weight	--	1.5	--	1.5	--	1.5
Distillation: The distillate percent by weight on the water-free basis shall be within the following limits:						
Up to 210° C.	--	2	--	2	--	2
Up to 235° C.	--	12	--	12	--	12
Up to 270° C.	20	40	20	40	20	40
Up to 315° C.	45	65	40	65	45	65
Up to 355° C.	75	82	65	82	65	75
Specific gravities of fractions:						
Fraction 235° to 315° C. at 38° C. compared with water at 15.5° C.	1.027	--	1.027	--	1.030	--
Fraction 315° to 355° C. at 38° C. compared with water at 15.5° C.	1.095	--	1.095	--	1.110	--
Fraction above 355° C. (distillation residue at 38° C. compared with water at 15.5° C.)	--	--	--	--	1.160	--

1/ These requirements apply to creosote in use at treating plants and are not subject to any exceptions or tolerances for changes taking place during treating operations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling shall be conducted in accordance with AWPA Standard A4. The treating material shall be thoroughly agitated after which a 1-quart sample shall be taken for inspection and test purposes. The contractor shall supply such samples and containers as may be required at no cost to the Government.

4.3 Test methods.

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4.3.1 Properties. Tests to determine the properties of coal-tar creosote (see 3.2 and table I) shall be made in accordance with AWPA Standard A1.

4.4 Inspection of preparation for delivery. The samples selected in accordance with 4.2 shall be examined to determine compliance with section 5, and as required by specifications referenced therein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. Unless otherwise specified in the contract or order, creosote shall be packaged in five-gallon pails conforming to PPP-P-704, type I, class 1 or 8, except commercial exterior coatings are acceptable for level B. Pails shall be furnished with pour spouts (see 6.2).

5.1.2 Level C. The filled five-gallon pails shall be packaged to provide adequate protection against corrosion, deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B.

5.2.1.1 Five-gallon pails. Unless otherwise specified in the contract or order (see 6.2), no overpacking of filled five-gallon pails shall be required. When specified, filled five-gallon pails shall be palletized in accordance with the requirements for load type III of MIL-STD-147.

5.2.1.2 Fifty-five gallon drums. Fifty-five gallons of creosote shall be packed in nominal 55-gallon drums conforming to type I or II of PPP-D-729, as specified (see 6.2).

5.2.1.3 Bulk. Tank-car shipments of creosote shall be prepared in accordance with the Interstate Commerce Commission Regulations.

5.2.2 Level C. Cresote, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in accordance with the Interstate Commerce Commission Regulations, Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking and labeling.

5.3.1 Labeling. All shipping containers shall be labeled in accordance with applicable Interstate Commerce Commission Regulations. The labeling shall be registered under the Federal Insecticide, Fungicide, and Rodenticide Act, as amended, in compliance with its requirement. Unless otherwise specified, each individual container shall be plainly marked with the following information:

Creosote, Coal Tar Technical.
Federal Specification TT-C-645b.
Class.
Quantity (gallons at 60° F.).
Gross weight.
Name of manufacturer (and name of contractor if not the same).
Government order number (or contract number if no order number is assigned).

CAUTION:

May cause skin irritation.
Avoid prolonged or repeated contact with skin.
Avoid prolonged breathing of vapor.
Use only with adequate ventilation.

5.3.2 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.3 Military agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

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6.1 Intended use. The intended uses, as described in Federal Specification TT-W-571, Wood Preservation: Treating Practices, are as follows:

Class 1 - For pressure treatment of poles.
Class 2 - For pressure treatment of ties, lumber, structural timbers, land piles, and posts, or for butt-treating poles.
Class 3 - For pressure treatment of piles, lumber, and structural timbers all used in coastal waters.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Class required (see 1.2.1 and 6.1).
(c) Administrative provision for inspection records (see 4.1).
(d) Size of container required.
(e) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

MILITARY INTEREST:

Custodians:

Army - ME
Navy - YD
Air Force - 68

Review activities:

Army - CE, MD, MU
Navy - YD
Air Force - 68

User activities:

Army - AT, SM
Navy - CG

Preparing activity:

AGR. - FS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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